Inference-time proofreader guidance

Can the released proofreader improve complete contact maps by changing sampling, looking ahead, maintaining several prefixes, repairing errors, or combining larger independent rollout pools?

Frozen generator: exp277 step 266344. Frozen proofreader: exp356 step 18000.
Implementation and GPU validation are in progress; no guided-policy result yet.

Frozen models and experimental references

The generator is the exp277 step-266344 Qwen3 model. The proofreader freezes its causal backbone and adds a four-layer, 512-wide bidirectional contact reader with 22.18 million trainable parameters. It predicts each supplied contact's correctness and the current rollout's recall. No model is retrained here.

The proofreader was trained on 50,000 experimental PDB chain structures. References use the complete exp222 native-only pyconfind contact set: a 3 Å side-chain threshold, minimum contact degree 0.001 and sequence separation ≥6. The validation split has 1,552 structures in 287 related groups.

What top ten percent means here

At each contact boundary, sample K candidate contacts from the constrained generator. Score them with the proofreader after appending each candidate. Keep the highest ceil(0.1 K) candidates and sample uniformly among survivors. This keeps 2 of 16 or 7 of 64 sampled proposals, including possible repeats; it does not enumerate the entire residue-pair universe.

Soft guidance weights the same proposals by q raised to a chosen power. Lookahead adds future contacts before scoring, then commits only the first. The generator's probability already enters through proposal sampling.

Complete nested-pool pilot

On 32 validation structures, sum-q R-precision increases from 0.5792 with eight rollouts to 0.5993 with 128. The paired gain is 0.0200, with 95% group-bootstrap interval 0.0050–0.0329. Long-range R-precision rises from 0.5537 to 0.5806.

Candidate coverage grows from 0.7765 to 0.9509. At 128 rollouts, frequency alone slightly exceeds sum-q: all-contact R-precision 0.6040, long-range 0.5864. Most remaining error is ranking error despite the much larger candidate pool.

Confidence tie-breaking and predicted-F1 vote weighting were added as exploratory validation tests. Their tiny gains over frequency have intervals including zero.

Baseline calibration and coverage

On 128 ordinary rollouts per pilot structure, per-rollout predicted precision has mean absolute error 0.0606 and predicted recall 0.0547. Errors average within structures and then over the same 32 structures. Contact calibration curves weight individual contacts; the two summaries answer different questions.

Mean trajectory overlap is 0.3473. Enlarging the pool exposes about 95% of native contacts, but ranking only recovers about 60% at the native contact-count budget. The remaining ranking gap motivates guidance and better aggregation; it does not establish that either can close the gap. Guided-rollout calibration is pending.

Practical single-batch baseline

Requesting each pool in one batch gives 0.6036 all-contact and 0.5894 long-range R-precision with frequency ranking at 128 rollouts. Mean warm generation cost is 7.165 H100 seconds per map; no proofreader pass is needed for this aggregator.

Against eight rollouts with sum-q ranking, the paired all-contact gain is 0.0244 (95% group interval 0.0101–0.0373) and the long-range gain is 0.0357 (0.0154–0.0519). The 128-rollout pool contains 95.39% of native contacts. The single-batch recipe is the practical cost baseline for guided search.

Larger validation confirms the pool-size benefit

On all 1,375 structures in 255 validation groups outside the pilot, frequency ranking improves from 0.5014 at eight rollouts to 0.5477 at 128. The paired gain is 0.0463 (95% group-bootstrap interval 0.0403–0.0526). Long-range R-precision rises from 0.4709 to 0.5215; its gain is 0.0506 (0.0430–0.0586).

The pool exposes 95.23% of native contacts. Mean warm generation time increases from 2.431 to 6.599 H100 seconds. Both sides use frequency ranking; fresh proofreader-weighted validation scoring remains pending. The long-range mean covers 1,371 structures with nonempty long-range references.

Benefit across lengths and measured latency

Increasing frequency-ranked pools from eight to 128 improves R-precision on 1,196 of 1,375 validation structures; 63 tie and 116 worsen. The median paired gain is 0.0392. Descriptive bands of 32–127, 128–255, 256–511 and 512–1000 residues all have positive mean all-contact and long-range gains.

For 128 rollouts, median warm generation latency is 4.345 H100 seconds, the 95th percentile is 17.024 seconds and the maximum is 63.331 seconds. Long proteins dominate the tail. Every plotted latency comes from its saved timing record; model loading and artifact writes are excluded from warm inference.

Exploratory length control

On the five short proteins completed by every original policy, top-10% filtering raises per-rollout precision but emits fewer contacts and lowers recall. This incomplete cohort motivates a stopping-rule ablation, not policy selection.

An added exploratory pilot compares ordinary and top-10%-of-16 sampling at exactly L and 2L contacts, with EOS disabled for both. Four independent 2L continuations yield nested L and 2L maps. The original 15-policy pilot remains frozen. No new reserved-test scores have been inspected.

Existing 1552-structure validation baseline

On existing validation rollouts, sum q remains strongest among five simple aggregators: all-contact R-precision 0.5315; long-range R-precision 0.5046. Frequency gives 0.5167 and 0.4877 respectively. Mean q alone gives 0.4895.

With up to eight rollouts, the pool contains about 75% of native contacts. This separates missing-contact generation from ranking errors. Oracle pool coverage is an upper bound, not the performance of an inference method.

Frozen pilot and execution

32 validation proteins from distinct homology groups span 33–937 residues. Compare top-percentile gates, soft weighting, lookahead, population search and one-pass repair, plus ordinary vLLM pools of 8, 32 and 128 rollouts.

Measure full-universe AP, all/long-range R-precision and complete search cost. Select policies on validation, then lock a reserved-test confirmation plan. Jobs use co-located CoreWeave storage and resumable per-rollout artifacts. H100 and GB200 execution phases retain separate timing provenance.

Execution and recoverability

A repeatedly preempted shared worker was canceled after repeated model staging exceeded its intended transfer budget. The incident is recorded. Recovery jobs use immutable source bundles, per-trajectory commit markers and durable S3. Regional model mirrors use persistent reservations with a 7 GB cap each.

The complete H100 ordinary baselines are saved. H100 guided and fixed-length runs were interrupted again at 18:15 UTC, and both H100 controllers stopped reporting. No shared controller, priority or other user's workload was changed. The separate GB200 ordinary phase passed full-forward scoring checks. Native H100 jobs now use the existing Kueue batch queue despite the Iris outage.

Numerical validation

All frozen backbone tensors match the generator exactly. CPU cache tests agree with full forward passes. The first H100 smoke caught BF16 incremental/full-pass drift of up to 0.02394 in one contact probability. FP32 cached search passes on three real proteins, with maximum error 0.00000358. Final maps use the released BF16 full-pass reader for comparison.

GB200 verification and its limits

Tiny-model cache checks and production ordinary generation passed on GB200. The production proofreader's CUDA 13 replay failed: the largest BF16 contact-score difference from saved H100 predictions was 0.0800, against a 0.01 guard. A real prefix also exceeded the unchanged 1e-5 FP32 cache-equivalence threshold.

Forcing math attention did not meet the BF16 guard; memory-efficient attention was unsupported for this cache path. Matching CUDA 12.8 passed the BF16 replay: two batches match exactly and the third has maximum difference 0.001523. One cache test still fails, so cached guidance remains pending. Ordinary generation and full-forward scoring can proceed. Cost plots reject mixtures of GPU models.

How to interpret the eventual comparison

R-precision evaluates the top R predictions, where R is the number of native contacts used only for evaluation. Precision at L, L/2 and L/5 provides usable sequence-length budgets. AP includes every eligible unproposed pair at score zero. Contact labels enter only after sampling decisions are finished.

Pilot and larger-validation results select the final methods. The reserved test was previously used for the proofreader release; new inference-policy test scores have not been inspected. Generator pretraining decontamination is not established. Search timings describe this implementation, including discarded work, and are not hardware-independent lower bounds on the cost of these policy families.

Stronger comparisons at larger compute budgets

A separate GB200 ordinary-generation phase requests 512 and 2,048 independent rollouts per pilot structure. These sizes were frozen before complete guided results were available. Observed lookahead costs already greatly exceeded the 128-rollout baseline, motivating a wider ordinary-sampling cost curve.

Frequency ranking needs no proofreader pass. These are measured compute envelopes, not exact budget matching. The same 32 validation structures are used, and the reserved test remains untouched for new policy metrics.

Reranking helps the small pilot pools

Replacing frequency by sum q changes 15.37% of top-R slots with eight rollouts and improves R-precision by 0.0163 (paired 95% interval 0.0090–0.0248). With 128 rollouts, it changes 7.62% of slots; the net effect is −0.0031 (−0.0099–0.0039), which does not establish a difference.

The diagnostic counts true contacts entering and leaving the map. Those counts exactly reproduce the observed metric difference for every protein. All 32 pilot structures are included; these results do not replace the larger validation test.

H100 execution restored without a controller restart

Kubernetes showed 320 unallocated H100s and a healthy existing batch queue. Native jobs were admitted through cw-rno2a-lq / iris-cq at the original batch priority. The real-model smoke passed, with maximum cache error 8.94e-7. All 22 stranded East replacements were canceled.

The saved guided campaign now resumes on 48 H100 workers. Original model, source, plans, per-trajectory seeds and saved predictions remain unchanged; only execution partitioning changes. Four fixed-length workers also resumed. Native job IDs and exact specifications are retained separately from Iris IDs.

Separate ordinary-pool replication on GB200

All 32 pilot proteins are complete for independent pools of 8, 32 and 128. Eight-rollout sum q gives all-contact R-precision 0.5783 and long-range 0.5591. At 128 rollouts, frequency gives 0.6025 and 0.5860; sum q gives 0.5995 and 0.5846.

Mean warm times are 1.511 seconds for eight-rollout sum q, 3.471 seconds for 128-rollout frequency, and 5.204 seconds for 128-rollout sum q. These use vLLM 0.14.1 rather than the H100 baseline's 0.9.2; this is a separate runtime cohort and not a controlled measurement of hardware speed alone.

Larger pools reach a frequency-ranking plateau

The complete GB200 pilot gives frequency R-precision 0.60248, 0.60319 and 0.60257 at 128, 512 and 2,048 rollouts. Native-contact coverage rises from 95.20% to 98.60% and 99.33%, while long-range R-precision is essentially flat.

The paired 2,048-minus-128 all-contact difference is +0.00009 (95% interval −0.00426 to 0.00503); long-range difference is −0.00148 (−0.00687 to 0.00411). Mean warm cost rises from 3.471 to 82.198 GB200 seconds, a factor of 23.68. This does not establish an accuracy gain. Proofreader ranking at 2,048 is untested.

Larger validation replication on GB200

All 1,375 structures from 255 groups are complete for frequency ranking. Eight versus 128 rollouts gives R-precision 0.50086 versus 0.54779; the paired gain is 0.04693 (95% interval 0.04052 to 0.05353). Long-range R-precision rises from 0.47099 to 0.52194, a gain of 0.05096 (0.04338 to 0.05915).

Mean warm cost is 1.196 versus 3.054 GB200 seconds. This separate runtime replicates the H100 frequency result. Full-forward proofreader scoring is also complete. No new reserved-test scores have been computed or inspected.

Complete H100 validation: confidence helps small pools

The released reader has scored all 2,750 pools from 1,375 validation structures. At eight rollouts, sum q improves R-precision from 0.50137 to 0.51737: paired gain 0.01601 (95% interval 0.01239 to 0.02033). Long-range gain is 0.01710 (0.01240 to 0.02223), at 1.061 times the measured warm inference cost.

At 128 rollouts, sum q and frequency are effectively tied in this comparison: all-contact difference −0.00008 (−0.00386 to 0.00424), long-range +0.00089 (−0.00371 to 0.00599). Mean confidence alone loses the useful vote-count signal.

Rollout quality weighting gives a small validation gain

With 128 rollouts, weighting each trajectory's votes by its predicted F1 gives R-precision 0.55195 and long-range R-precision 0.52565. Compared with frequency, the paired gains are 0.00426 (95% interval 0.00173 to 0.00763) and 0.00412 (0.00129 to 0.00781). Mean warm H100 time is 8.921 versus 6.599 seconds.

The weight uses predicted precision and recall only. This aggregation rule was added after the pilot; intervals are not adjusted for method selection. It is a promising validation result, not a reserved-test confirmation. Guided-policy selection still waits for the complete pilot.

Calibration and contact swaps explain the aggregation patter

On the full H100 panel, sum q replaces 18.29% of top-R slots at eight rollouts and 8.70% at 128. Correct contacts gained minus correct contacts lost exactly reproduce each protein's R-precision change. At 128 the two nearly cancel.

The 128-rollout precision and recall prediction MAEs are 0.05238 and 0.05768, averaged within each structure then across structures. Mean predicted recall is 0.38840 versus observed 0.40864. Good pooled contact calibration does not by itself guarantee accurate rollout recall or improved map ranking.

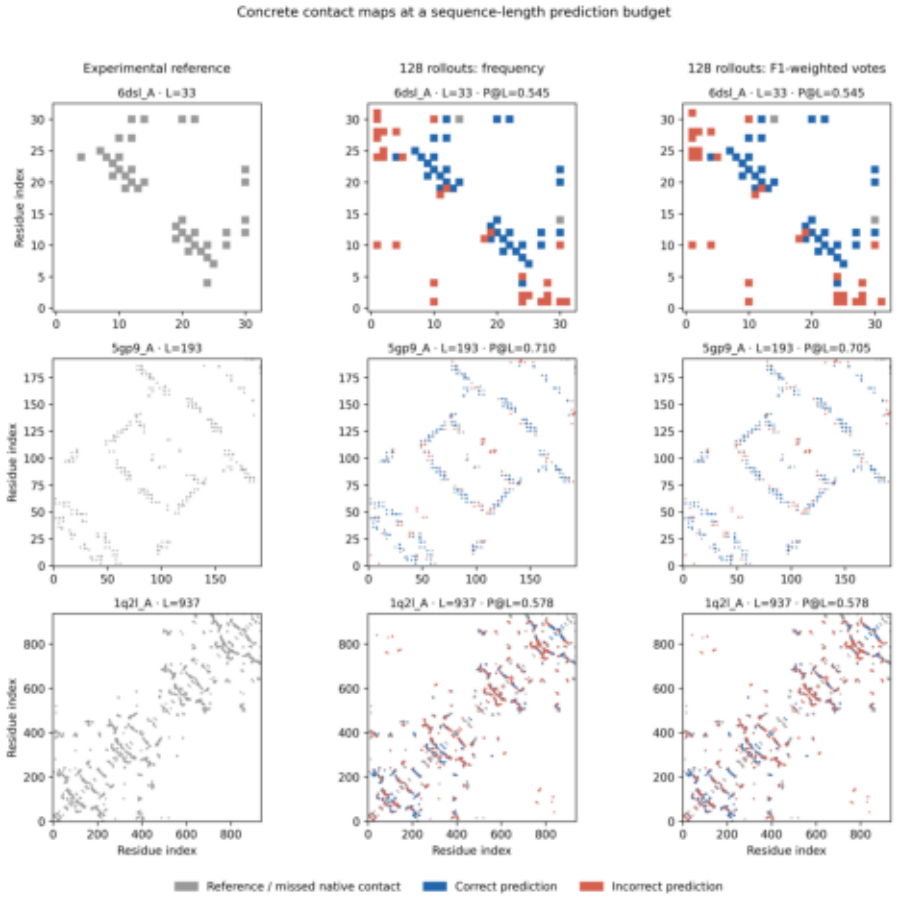
GB200 replicates the rollout-F1 weighting gain

All 2,750 GB200 validation pools are scored. At 128 rollouts, F1-weighted votes give R-precision 0.55239 and long-range R-precision 0.52647. Paired gains over frequency are 0.00459 (95% interval 0.00207 to 0.00790) and 0.00453 (0.00164 to 0.00813). Sum q again has no established advantage at this pool size.

Mean warm generation-plus-proofreading time is 4.622 GB200 seconds, versus 3.054 for frequency alone. These are the same validation proteins as the H100 phase, so this is a runtime/stochastic replication, not a new biological cohort.

contact_examples.png

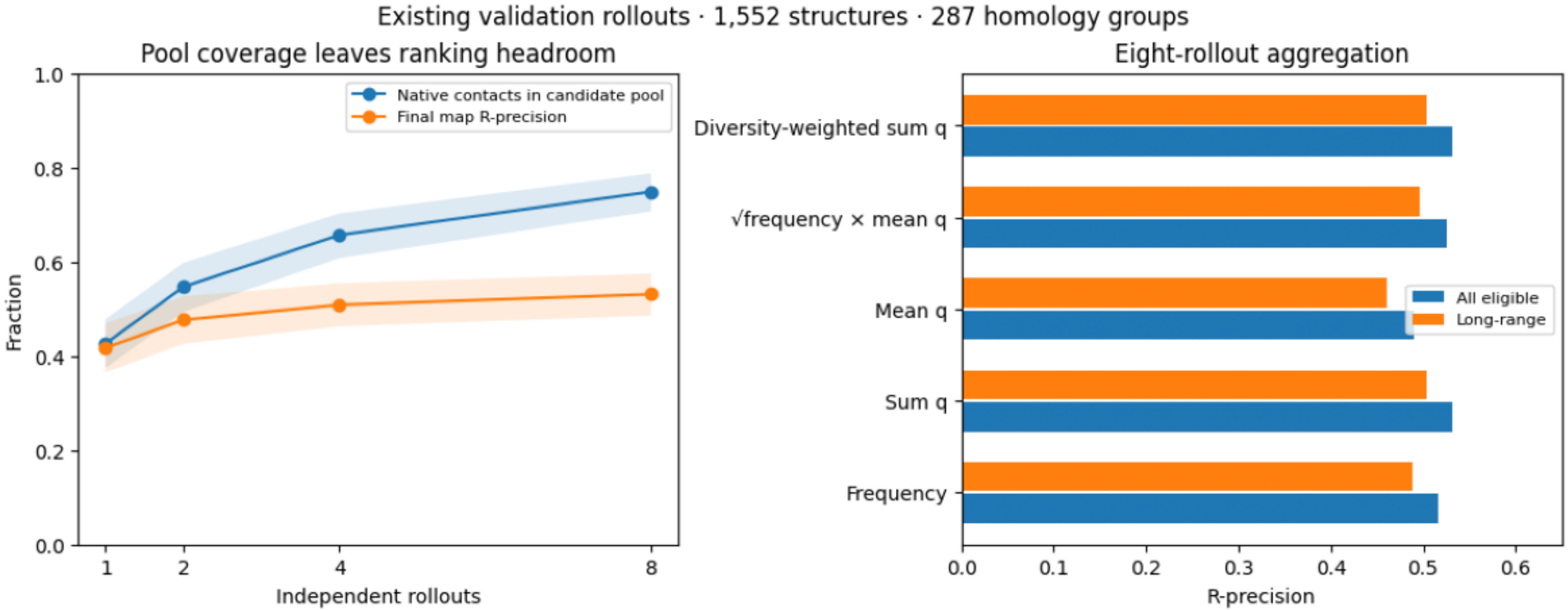
Three validation proteins chosen as the first, middle and last length ranks for earlier numerical checks, without accuracy selection. Both methods rank the same 128 H100 rollouts and predict L unique pairs. Matrices are mirrored only for visualization; each pair counts once. Gray native contacts remain visible where not selected. Point markers are enlarged for legibility on long sequences. These examples are descriptive and do not estimate an average effect. References use the inherited exp222 native-only pyconfind definition, with sequence separation at least six. The published coordinate CSV reproduces this figure.




```
$ plot_contact_examples.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/contact_examples.csv.gz --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/contact_examples.png
```

existing_validation.png

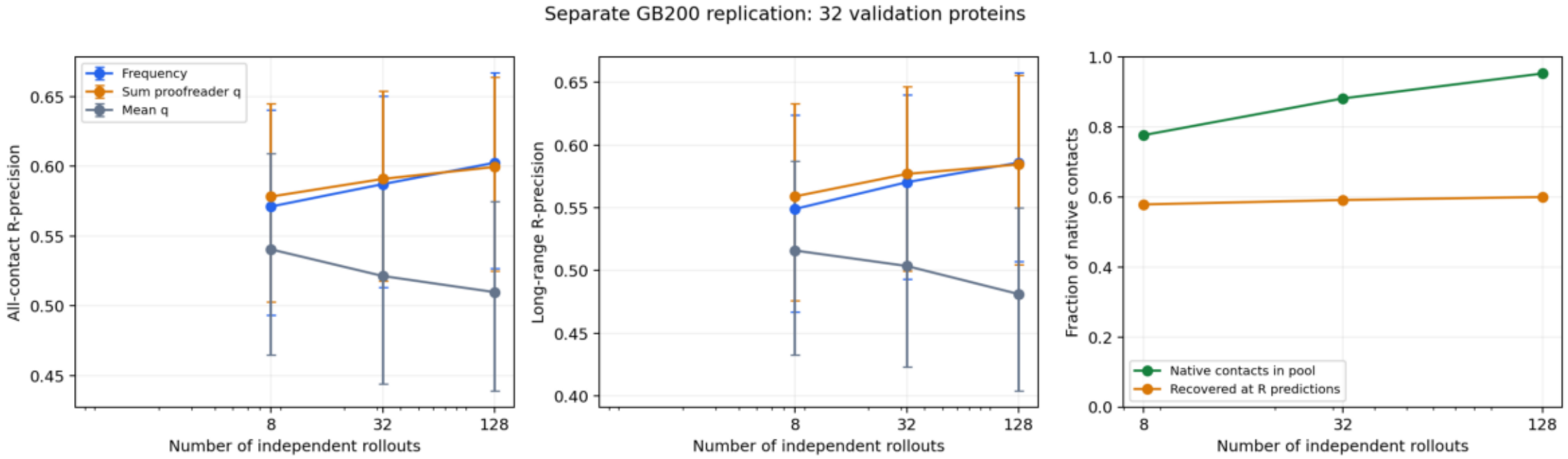
Validation-only reanalysis. Bands resample homology groups (95% intervals). Oracle coverage uses labels only to bound achievable recall. The small number of structures with fewer than eight usable rollouts retains all available samples. Long-range means separation ≥24.



$ plot_existing.py

pilot_direct_gb200.png

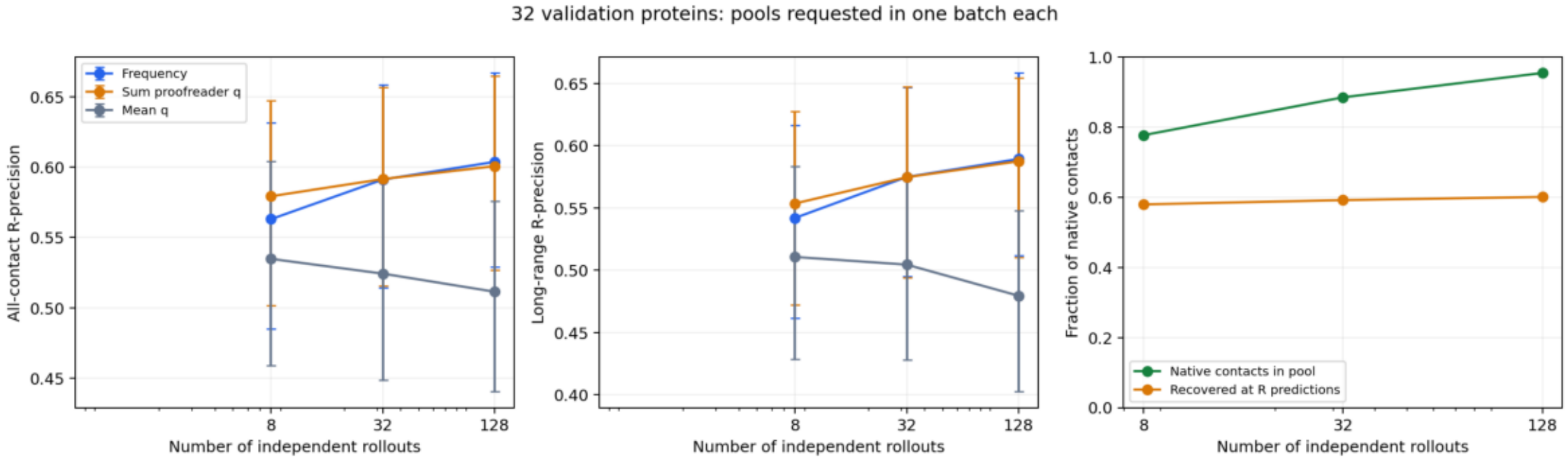
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_direct_gb200 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_direct_gb200.png --title 'Separate GB200 replication: 32 validation proteins'
```

pilot_direct_pool.png

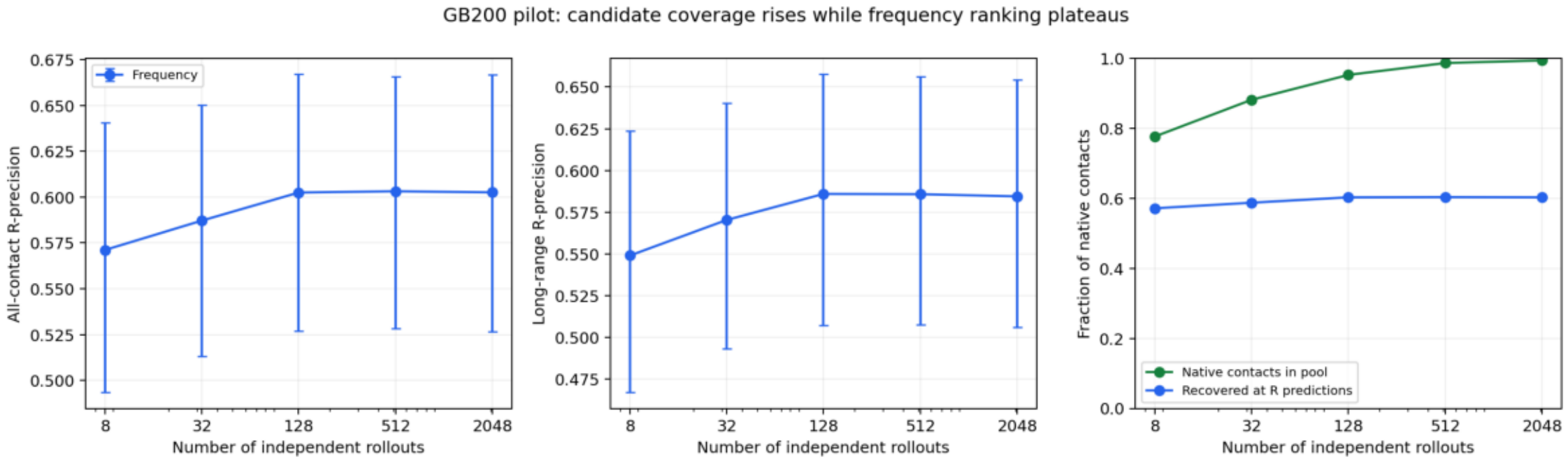
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_direct --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_direct_pool.png --title '32 validation proteins: pools requested in one batch each'
```

pilot_large_gb200.png

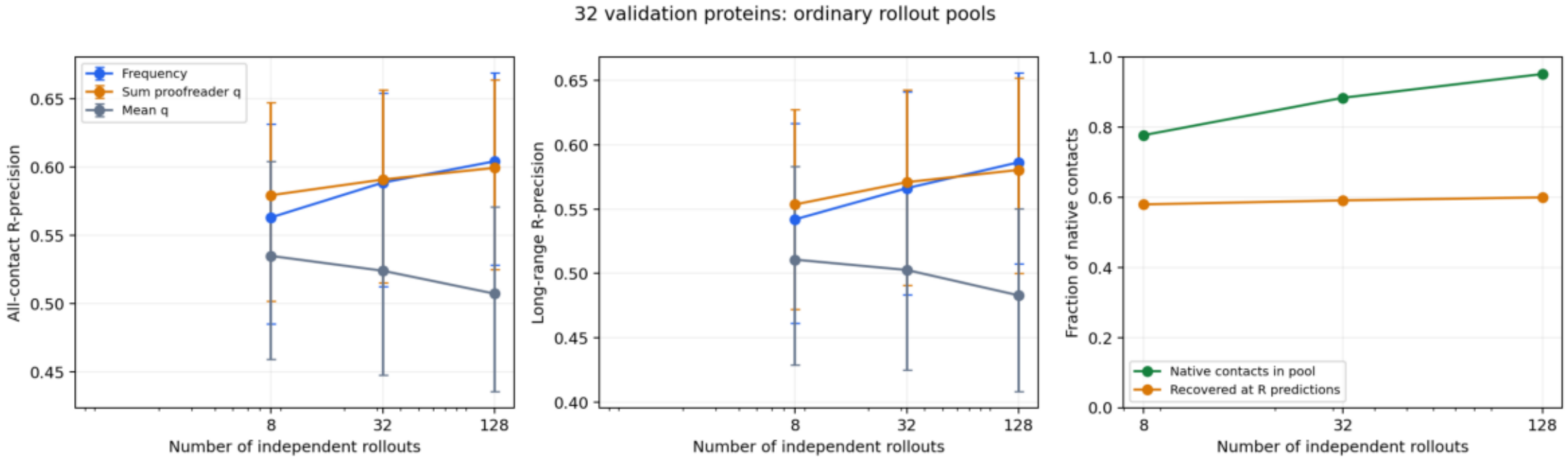
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool_scaling_gb200 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_large_gb200.png --methods frequency --coverage-method frequency --title \
  'GB200 pilot: candidate coverage rises while frequency ranking plateaus'
```

pilot_pool.png

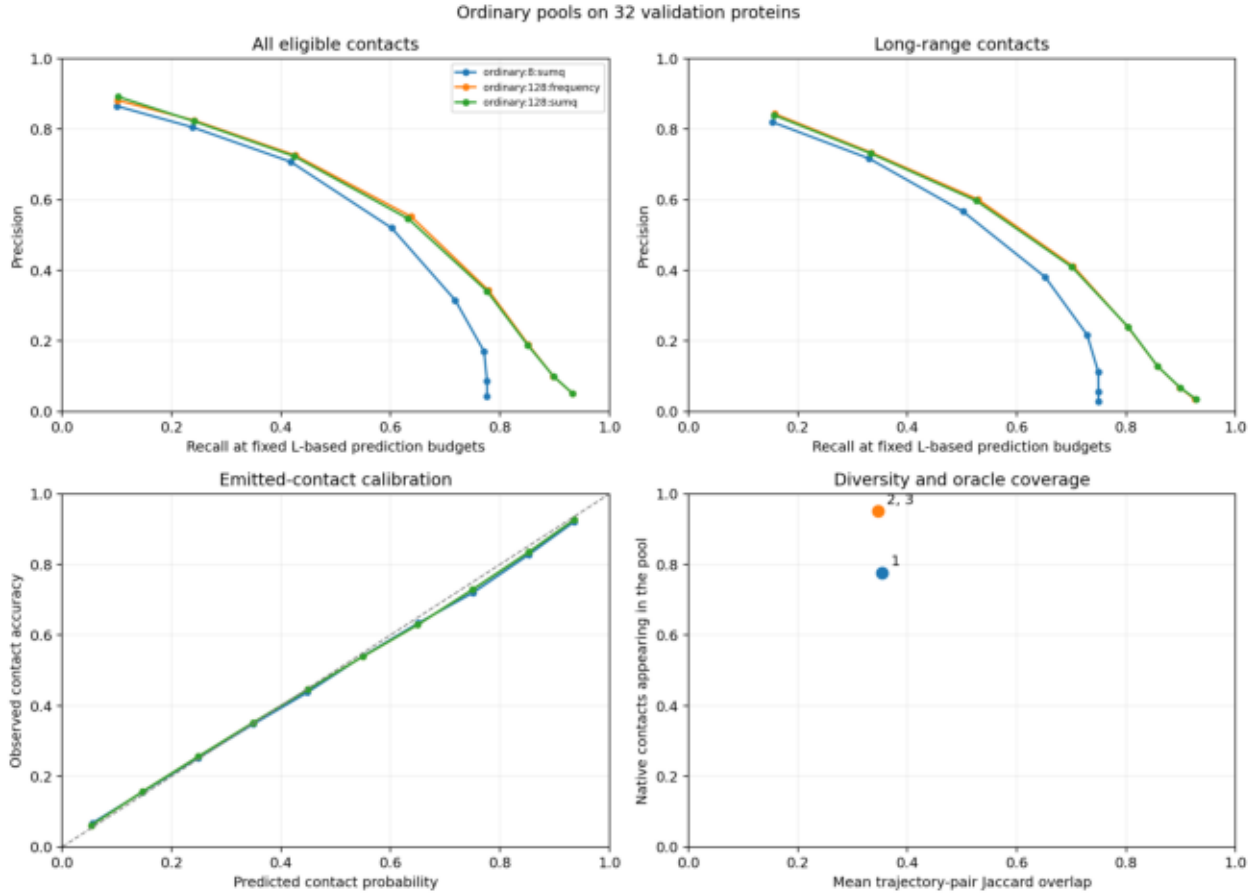
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_pool.png
```

pilot_pool_diagnostics.png

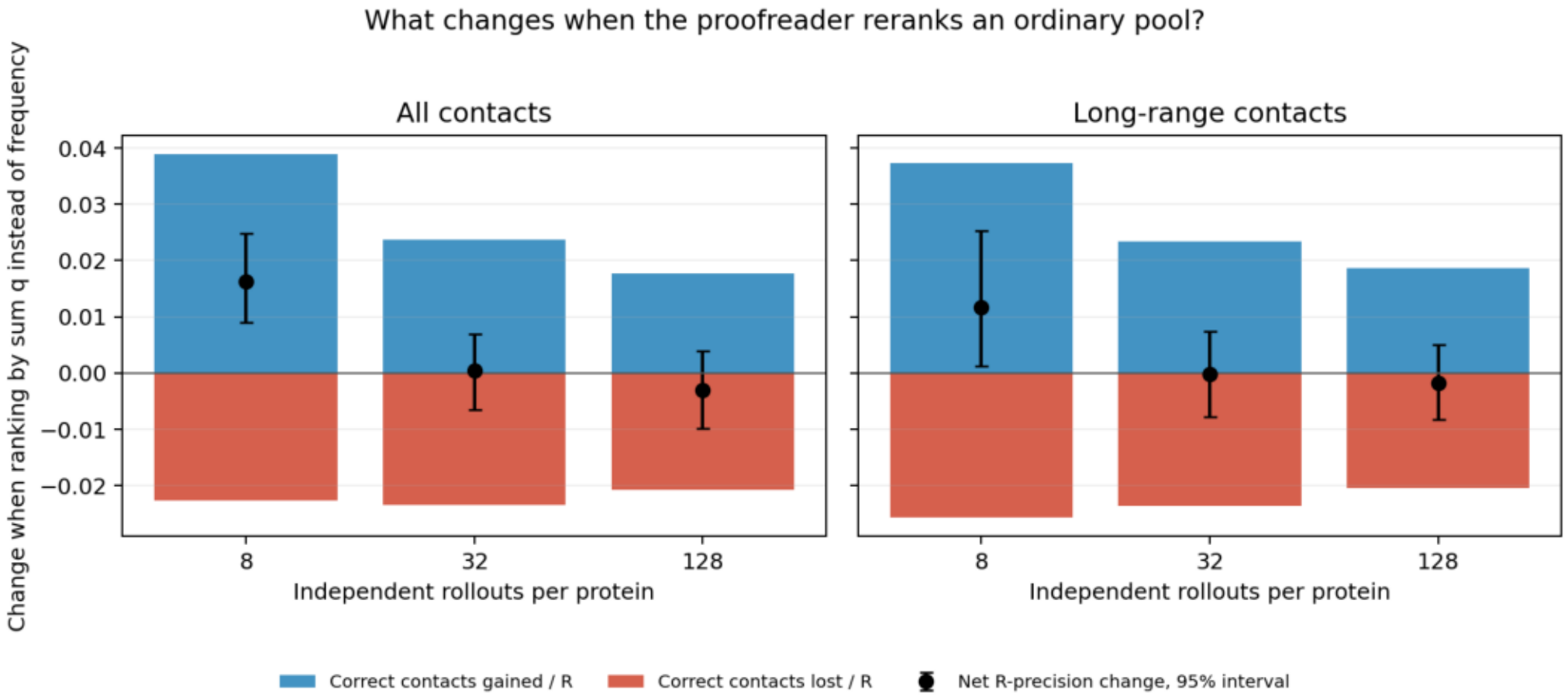
Curves average over the same structures at budgets 0.1, 0.25, 0.5, 1, 2, 4, 8 and 16 times sequence length. Unfilled slots count as misses; curves do not use native contact counts to choose the prediction budget. Calibration pools first occurrences within each rollout and is contact-weighted. Diversity and coverage are structure means; at most 128 deterministically sampled trajectory pairs measure overlap. Numbers in the coverage panel follow legend order; multiple aggregators of the same pool share calibration and diversity.




```
$ plot_diagnostics.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool/diagnostics --spec ordinary:8:sumq ordinary:128:frequency ordinary:128:sumq \
  --title 'Ordinary pools on 32 validation proteins' --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_pool_diagnostics.png
```

pilot_reranking.png

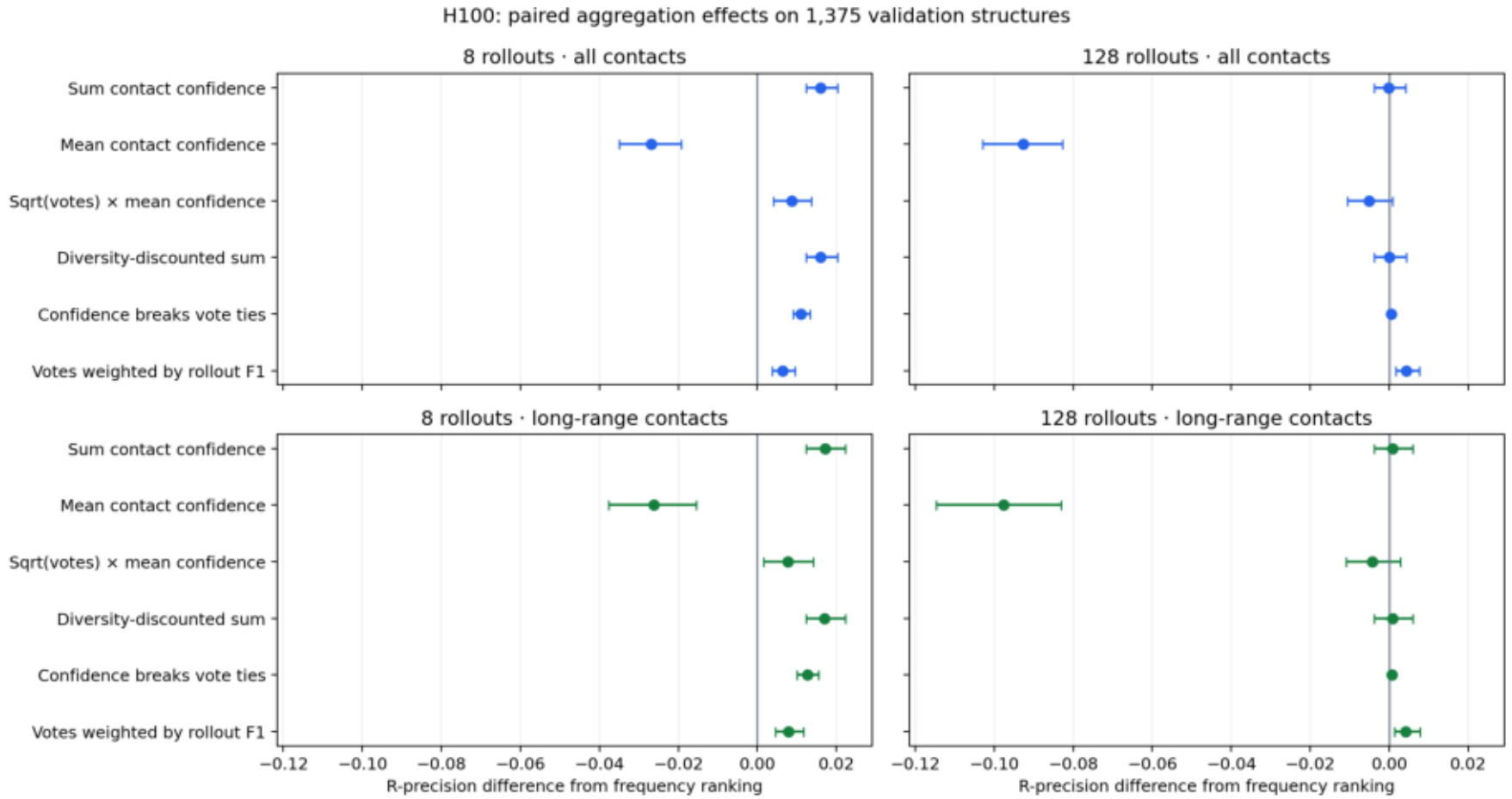
Both rankings use exactly the same sampled candidate union and deterministic label-independent tie breaks. Bars count true contacts entering and leaving the top R; their sum equals the observed R-precision change, verified for every input. Proteins are weighted equally; intervals resample homology groups. R uses native contact count only for retrospective evaluation. This is an adaptive diagnostic, not a new policy.




```
$ analyze_reranking.py --root experiments/exp368_evals_proofreader_guided_contact_maps/_cache/pilot-direct-scored --plan \
  experiments/exp368_evals_proofreader_guided_contact_maps/pilot_direct.json --targets experiments/exp356_models_pdb_rollout_proofreader/_cache/prepared-v2/targets.parquet --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_direct --plot experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_reranking.png
```

validation_aggregation_effects.png

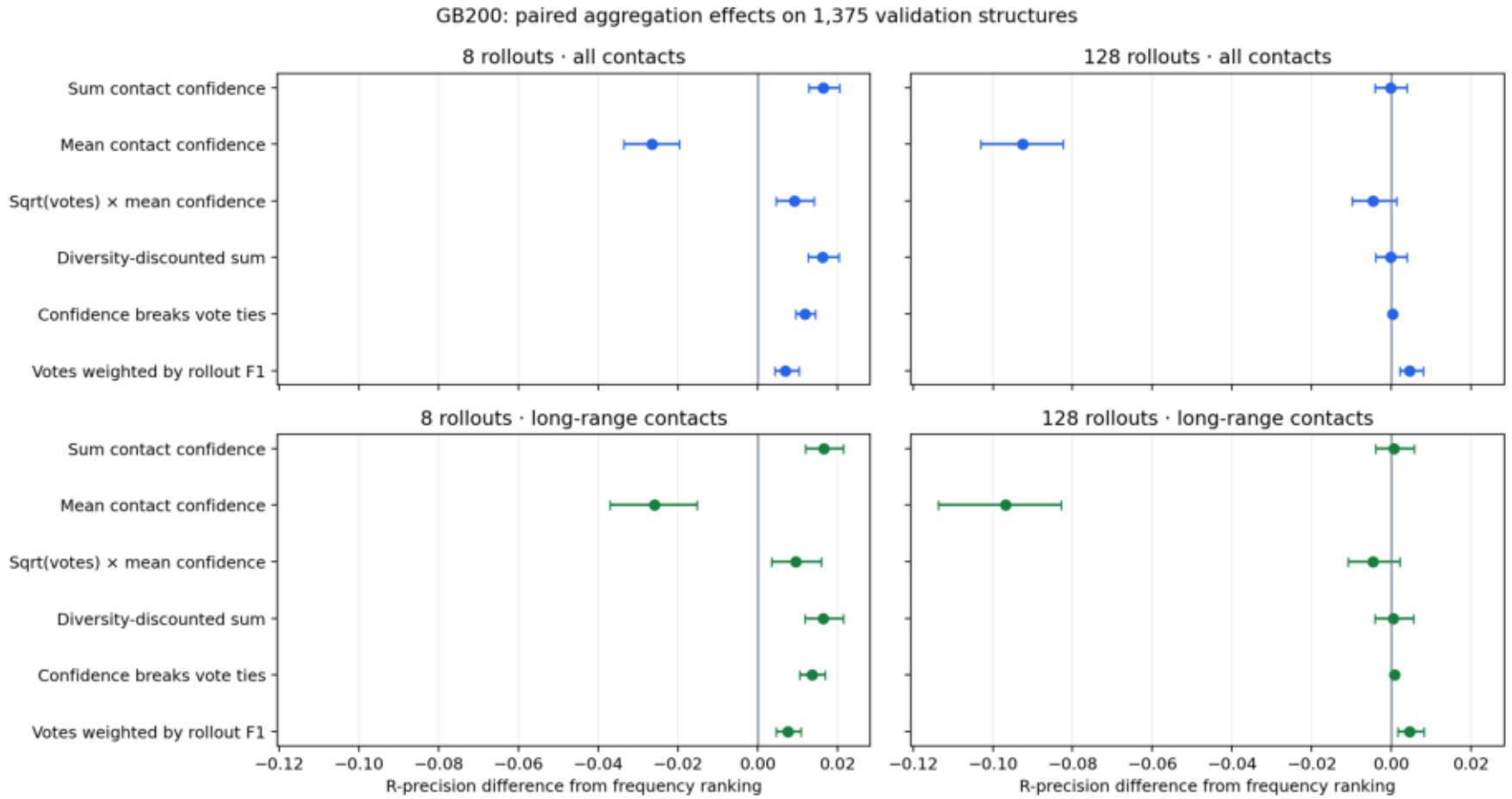
Paired differences on identical generated pools. Error bars are 95% homology-group bootstrap intervals, without multiplicity adjustment. All planned structures are included; long-range metrics exclude structures with no native contact at sequence separation ≥24. F1 weighting and confidence tie-breaking were added after the pilot and remain exploratory. No new reserved-test scores are shown.




```
$ plot_aggregation_effects.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_pool --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_aggregation_effects.png --title 'H100: paired aggregation effects on 1,375 validation structures'
```

validation_aggregation_effects_gb200.png

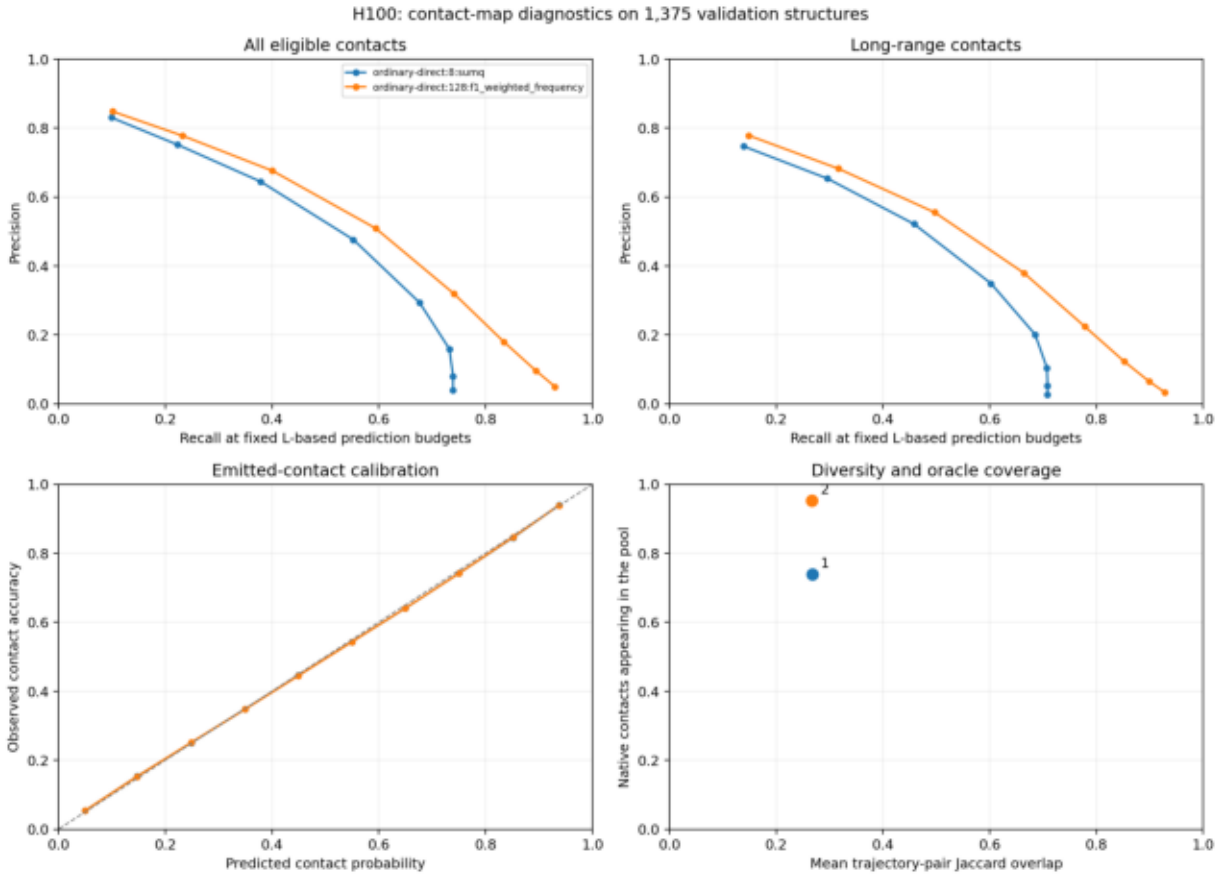
Paired differences on identical generated pools. Error bars are 95% homology-group bootstrap intervals, without multiplicity adjustment. All planned structures are included; long-range metrics exclude structures with no native contact at sequence separation ≥24. F1 weighting and confidence tie-breaking were added after the pilot and remain exploratory. No new reserved-test scores are shown.




```
$ plot_aggregation_effects.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_pool_gb200 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_aggregation_effects_gb200.png --title 'GB200: paired aggregation effects on 1,375 validation structures'
```

validation_diagnostics.png

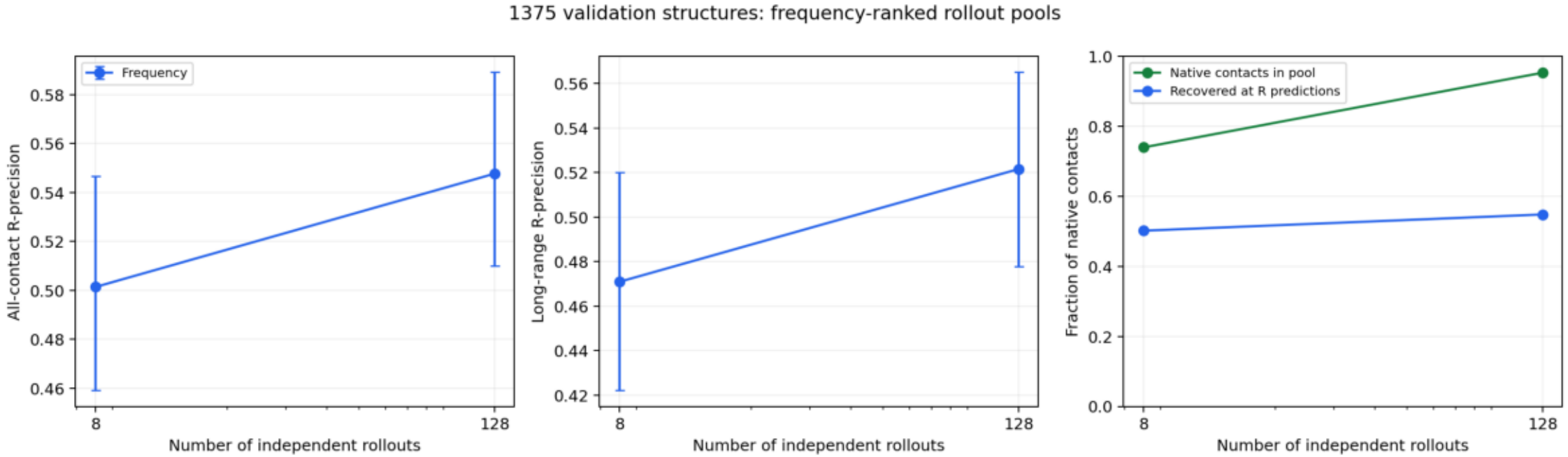
Curves average over the same structures at budgets 0.1, 0.25, 0.5, 1, 2, 4, 8 and 16 times sequence length. Unfilled slots count as misses; curves do not use native contact counts to choose the prediction budget. Calibration pools first occurrences within each rollout and is contact-weighted. Diversity and coverage are structure means; at most 128 deterministically sampled trajectory pairs measure overlap. Numbers in the coverage panel follow legend order; multiple aggregators of the same pool share calibration and diversity.




```
$ plot_diagnostics.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_diagnostics --spec ordinary-direct:8:sumq \
  ordinary-direct:128:f1_weighted_frequency --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_diagnostics.png --title \
  'H100: contact-map diagnostics on 1,375 validation structures'
```

validation_frequency.png

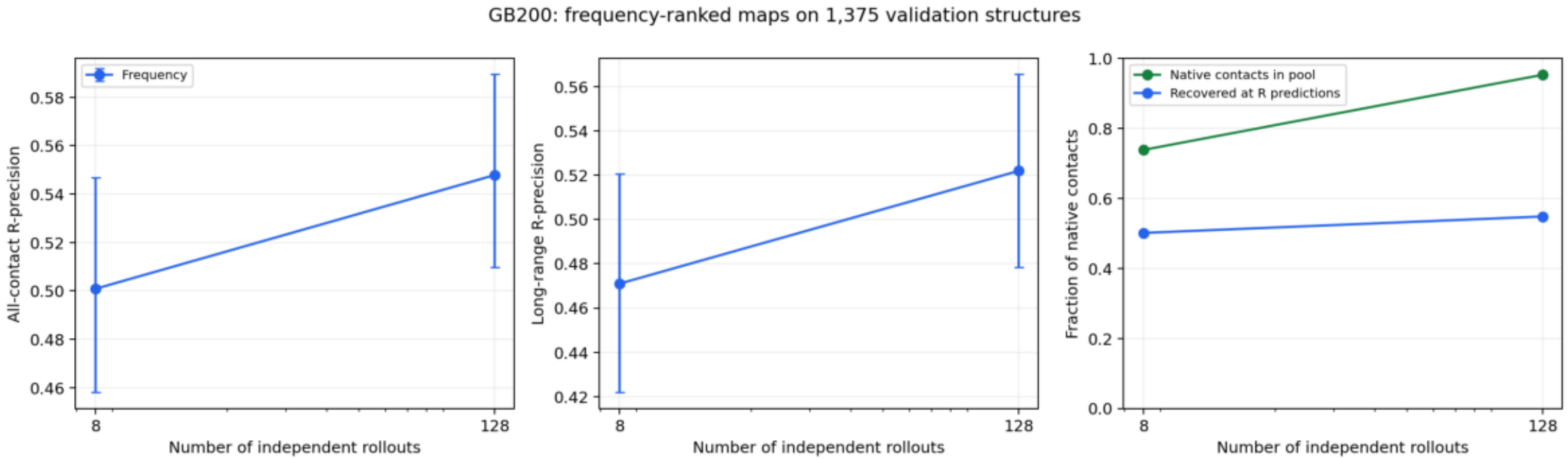
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_frequency.png --title '1375 validation structures: frequency-ranked rollout pools' --methods frequency \
  --coverage-method frequency
```

validation_frequency_gb200.png

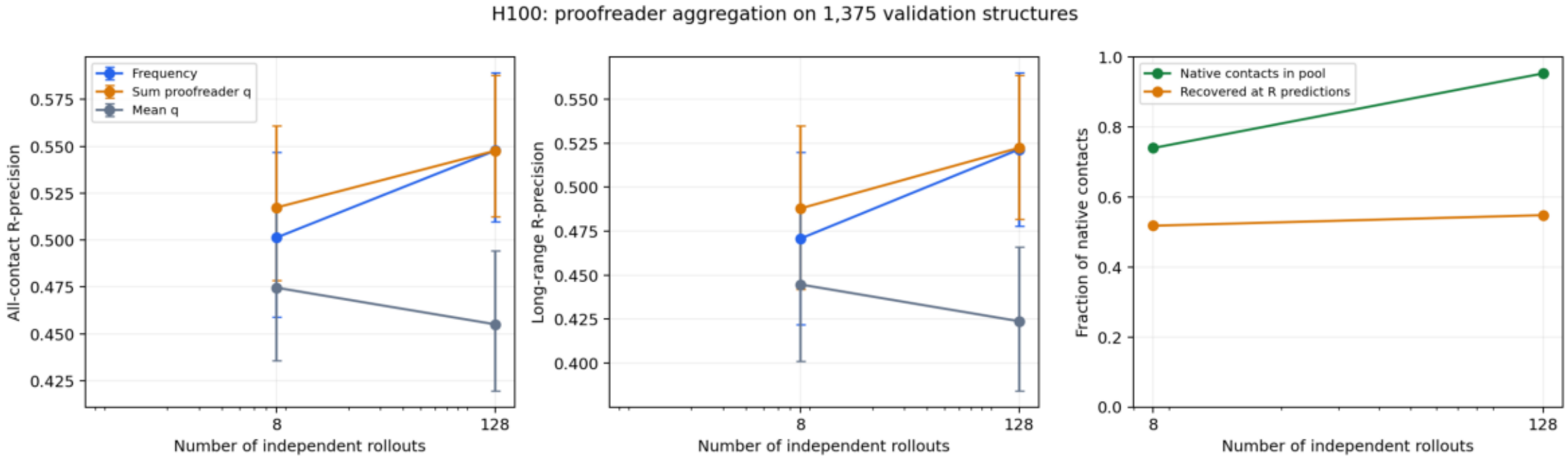
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency_gb200 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_frequency_gb200.png --title 'GB200: frequency-ranked maps on 1,375 validation structures' --methods \
  frequency --coverage-method frequency
```

validation_pool.png

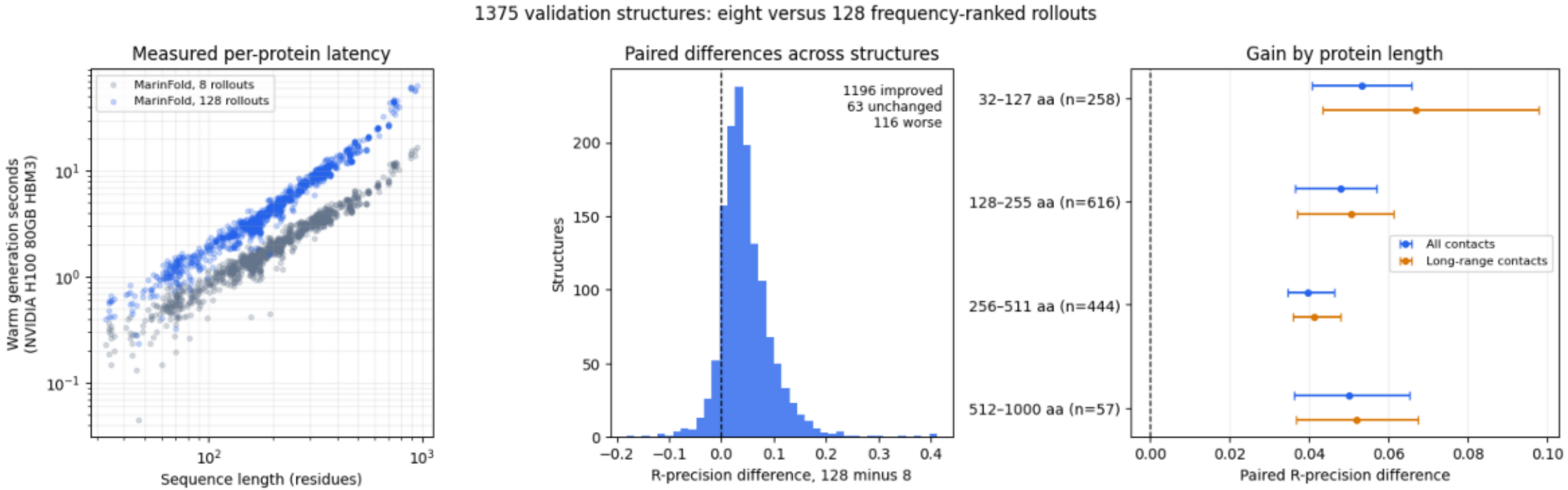
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_pool --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_pool.png --title 'H100: proofreader aggregation on 1,375 validation structures'
```

validation_pool_scaling.png

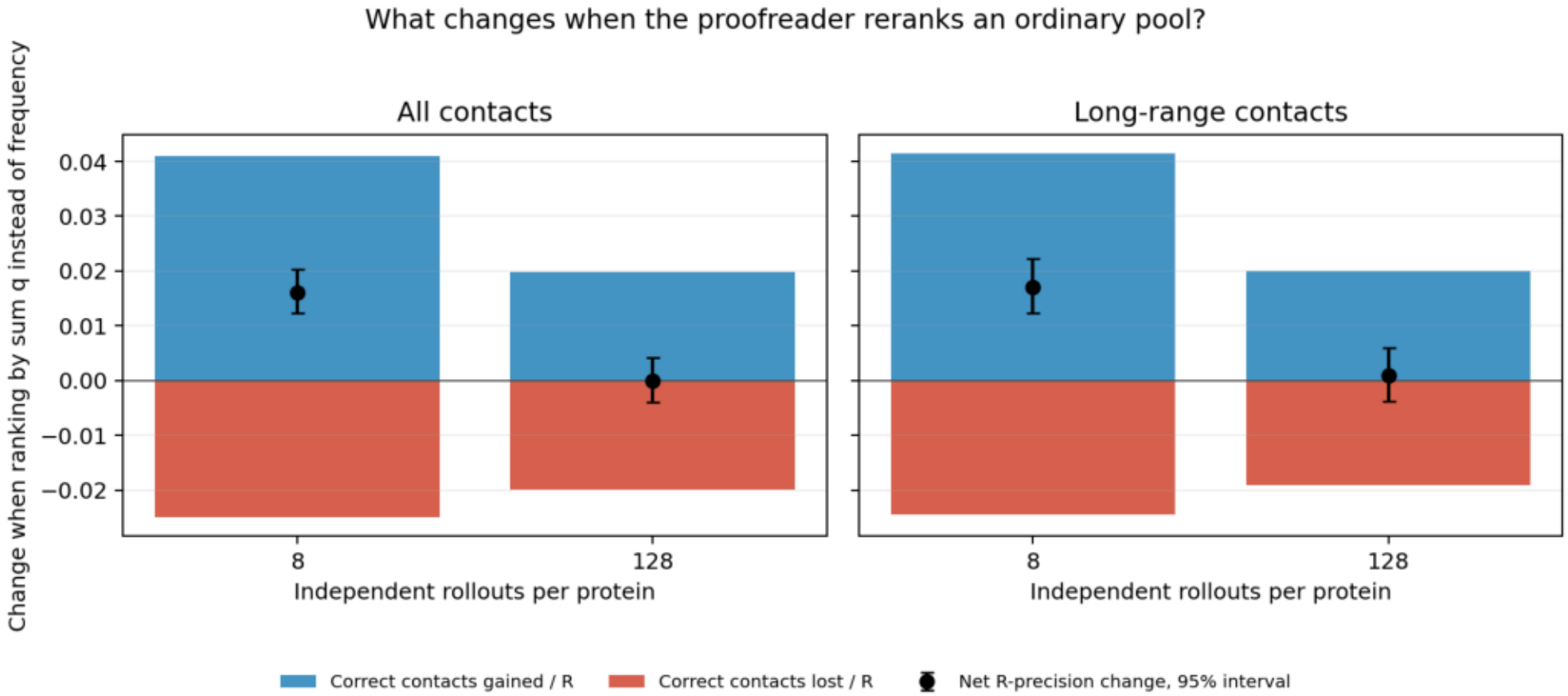
Each latency point is one (protein, pool-size) inference, excluding model load and artifact writes. Length strata use fixed boundaries at 32, 128, 256, 512 and 1001 residues. Intervals resample related groups within each stratum and describe paired differences. Strata are descriptive; they do not select proteins or tune a test policy.




```
$ plot_pool_scaling.py --input experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency/per_protein.csv.gz --plan \
  experiments/exp368_evals_proofreader_guided_contact_maps/validation_pool.json --data-out experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_pool_scaling.png
```

validation_reranking.png

Both rankings use exactly the same sampled candidate union and deterministic label-independent tie breaks. Bars count true contacts entering and leaving the top R; their sum equals the observed R-precision change, verified for every input. Proteins are weighted equally; intervals resample homology groups. R uses native contact count only for retrospective evaluation. This is an adaptive diagnostic, not a new policy.




```
$ analyze_reranking.py --plan experiments/exp368_evals_proofreader_guided_contact_maps/validation_pool.json --targets \
  experiments/exp356_models_pdb_rollout_proofreader/_cache/prepared-v2/targets.parquet --root \
  experiments/exp368_evals_proofreader_guided_contact_maps/_cache/validation-pool-scored-native-v1 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_pool --plot experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_reranking.png
```